

CHAPTER 23

SEISMIC DESIGN REFERENCE DOCUMENTS

23.1 CONSENSUS STANDARDS AND OTHER REFERENCE DOCUMENTS

This section lists the consensus standards and other documents that shall be considered part of this standard to the extent referenced in Chapters 11 to 22. Those referenced documents identified by an asterisk (*) are not consensus standards; rather, they are documents developed within the industry and represent acceptable procedures for design and construction to the extent referred to in the specified section.

***AAMA 501.6**, *Recommended Dynamic Test Method for Determining the Seismic Drift Causing Glass Fallout from a Wall System*, American Architectural Manufacturers Association, 2001.

Cited in: Section 13.5.9.2

ACI 307, *Code Requirements for Reinforced Concrete Chimneys and Commentary*, American Concrete Institute, 2008.

Cited in: Sections 15.4.1, 15.6.2.2, Table 15.4-2

ACI 313, *Standard Practice for Design and Construction of Concrete Silos and Stacking Tubes for Storing Granular Materials*, American Concrete Institute, 1997.

Cited in: Sections 15.7.9.3.3, 15.7.9.6, 15.7.9.7

ACI 318, *Building Code Requirements for Structural Concrete and Commentary*, American Concrete Institute, 2014.

Cited in: Sections 11.2, 12.12.5, 12.13.8, 12.13.9.2.1.1, 12.13.9.2.1.2, 12.13.9.3.3, 12.13.9.3.4, 13.4.2.1, 13.4.4, 13.5.7.2, 14.2.1, 14.2.2, 14.2.2.1, 14.2.2.2, 14.2.2.3, 14.2.2.4, 14.2.2.5, 14.2.2.6, 14.2.2.7, 14.2.3, 14.2.3.1.1, 14.2.3.2.1, 14.2.3.2.2, 14.2.3.2.3, 14.2.3.2.5, 14.2.3.2.6, 14.2.4, 14.2.4.5.5, 14.3.1, 14.3.2, 14.4.4.1.2, 14.4.5.2, 15.4.9.1, 15.4.9.4, 15.6.2.2, 15.7.5, 15.7.11.7, Table 12.2-1, Table 12.10-1, Table 15.4-1, Table 15.4-2

ACI 350.3, *Seismic Design of Liquid-Containing Concrete Structures and Commentary*, American Concrete Institute, 2006.

Cited in: Sections 15.7.6.1.1, 15.7.7.3

ACI 355.2, *Qualification of Post-Installed Mechanical Anchors in Concrete and Commentary*, American Concrete Institute, 2007.

Cited in: Sections 13.4.2.3, 15.4.9.3

ANSI/AHRI Standards 1270 (I-P) and 1271 (SI) Sections 1 through 10, excluding Appendices, *Requirements for Seismic Qualification of HVACR Equipment*, Air-Conditioning, Heating, and Refrigeration Institute, 2015.

Cited in: Section 13.6.2.1

ANSI/AISC 341, *Seismic Provisions for Structural Steel Buildings*, American Institute of Steel Construction, 2016.

Cited in: Sections 12.13.9.3.4, 14.1.1, 14.1.2.2.1, 14.1.2.2.2, 14.1.8, 14.3.1, 14.3.3, 17.2.5.5, Table 15.4-1

ANSI/AISC 360, *Specification for Structural Steel Buildings*, American Institute of Steel Construction, 2016.

Cited in: Sections 14.1.1, 14.1.2.1, 14.1.2.2.1, 14.3.1, 14.3.2, Table 15.4-1

ANSI/AISI S100, *North American Specification for the Design of Cold-Formed Steel Structural Members*, American Iron and Steel Institute, 2016.

Cited in: Sections 14.1.1, 14.1.3.1, 14.1.3.2, 14.1.4.1, 14.1.4.2.1, 14.1.5, Table 15.4-1

ANSI/AISI S230, *Standard for Cold-Formed Steel Framing—Prescriptive Method for One- and Two-Family Dwellings*, American Iron and Steel Institute, 2015.

Cited in: Sections 14.1.1, 14.1.4.3

ANSI/AISI S310, *North American Standard for the Design of Profiled Steel Diaphragm Panels*, American Iron and Steel Institute, 2016.

Cited in: Sections 14.1.1, 14.1.5

ANSI/AISI S400, *North American Standard for Seismic Design of Cold-Formed Steel Structural Systems*, American Iron and Steel Institute, 2015.

Cited in: Sections 12.14.7.2, Table 12.2-1, 14.1.1, 14.1.3.2, 14.1.4.2.1, 14.1.4.2.2

ANSI/RMI MH 16.1, *Specification for the Design, Testing, and Utilization of Industrial Steel Storage Racks*, Rack Manufacturers Institute, 2012.

Cited in: Sections 15.5.3.1, 15.5.3.1.1, 15.5.3.1.2, 15.5.3.3

ANSI/RMI MH 16.3 *Specification for the Design, Testing, and Utilization of Industrial Steel Cantilevered Storage Racks*, Rack Manufacturers Institute, 2016.

Cited in: Section 15.5.3.2, 15.5.3.2.1, 15.5.3.3

ANSI/SDI-C, *Standard for Composite Steel Floor Deck-Slabs*, Steel Deck Institute, 2011.

Cited in: Sections 14.1.1, 14.1.5

ANSI/SDI-NC, *Standard for Non-Composite Steel Floor Deck*, Steel Deck Institute, 2010.

Cited in: Sections 14.1.1, 14.1.5

ANSI/SDI-QA/QC, *Standard for Quality Control and Quality Assurance for Installation of Steel Deck*, Steel Deck Institute, 2011.

Cited in: Sections 14.1.1, 14.1.5

ANSI/SDI-RD, *Standard for Steel Roof Deck*, Steel Deck Institute, 2010.

Cited in: Sections 14.1.1, 14.1.5

ANSI/SJI-CJ, *Standard Specification for Composite Steel Joists*, CJ-series, Steel Joist Institute, 2010.

Cited in: Section 14.1.1, 14.1.6

ANSI/SJI-100, *Standard Specification for K-Series, LH-Series and DLH-Series Open Web Steel Joists and for Joist Girders*, Steel Joist Institute, 2015.

Cited in: Sections 14.1.1, 14.1.6

API 12B, *Specification for Bolted Tanks for Storage of Production Liquids, 12B*, 16th Ed., American Petroleum Institute, 2014.

Cited in: Section 15.7.8.2

API 620, *Design and Construction of Large, Welded, Low-Pressure Storage Tanks*, 12th Ed., Addendum 1, American Petroleum Institute, 2014.

Cited in: Sections 15.4.1, 15.7.8.1, 15.7.13.1

API 650, *Welded Tanks for Oil Storage*, 12th Ed., Addendum 1, American Petroleum Institute, 2014.

Cited in: Sections 15.4.1, 15.7.8.1, 15.7.9.4

API 653, *Tank Inspection, Repair, Alteration, and Reconstruction*, 5th Ed., American Petroleum Institute, 2014.

Cited in: Section 15.7.6.1.9, Table 15.4-1

ASCE 4, *Seismic Analysis of Safety-Related Nuclear Structures and Commentary*, American Society of Civil Engineers, 1998.

Cited in: Section 12.9.1.3

ASCE 8, *Specification for the Design of Cold-Formed Stainless Steel Structural Members*, American Society of Civil Engineers, 2002.

Cited in: Sections 14.1.1, 14.1.3.1, 14.1.3.2, 14.1.5

ASCE 19, *Structural Applications of Steel Cables for Buildings*, American Society of Civil Engineers, 2010.

Cited in: Sections 14.1.1, 14.1.7

ASCE 41, *Seismic Evaluation and Retrofit of Existing Buildings*, American Society of Civil Engineers, 2013.

Cited in: Chapter 16

ASME A17.1, *Safety Code for Elevators and Escalators*, American Society of Mechanical Engineers, 2007.

Cited in: Sections 13.6.11, 13.6.11.3

ASME B31, *Standards of Pressure Piping*, ASME–American Society of Mechanical Engineers (consists of the following standards):

ASME B31.1, *Power Piping*, 2014.

ASME B31.3, *Process Piping*, 2014.

ASME B31.4, *Pipeline Transportation Systems for Liquids and Slurries*, 2012

ASME B31.5, *Refrigeration Piping and Heat Transfer Components*, 2013.

ASME B31.8, *Gas Transmission and Distribution Piping Systems*, 2014.

ASME B31.9, *Building Services Piping*, 2014.

ASME B31.11, *Slurry Transportation Piping Systems*, 2002.

ASME B31.12, *Hydrogen Piping and Pipelines*, 2014.

ASME B31EA-2010, *Standard for the Seismic Design and Retrofit of Above-Ground Piping Systems*, Addendum A, 2010.

Cited in: Sections 13.6.4.1, 13.6.7.1, Table 13.6-1

ASME BPVC, *Boiler and Pressure Vessel Code*, American Society of Mechanical Engineers (consists of the following codes):

BPVC-I, *Rules for Construction of Power Boilers*, 2015.

BPVC-IV, *Rules for Construction of Heating Boilers*, 2015.

BPVC-VIII Division 1, *Rules for Construction of Pressure Vessels*, 2015.

BPVC-VIII Division 2, *Rules for Construction of Pressure Vessels*, Alternative Rules, 2015.

BPVC-VIII Division 3, *Rules for Construction of Pressure Vessels*, Alternative Rules for Construction of High Pressure Vessels, 2015.

Cited in: Sections 13.6.10, 13.6.13, 15.7.11.2, 15.7.11.6, 15.7.12.2

ASME STS-1, *Steel Stacks*, American Society of Mechanical Engineers, 2011

Cited in: Section 15.6.2.3, Table 15.4-2

ASTM A108, *Standard Specification for Steel Bar, Carbon and Alloy, Cold-Finished*, ASTM International, 2007.

Cited in: Section 11.3

ASTM A307, *Standard Specification for Carbon Steel Bolts and Studs, 60 000 PSI Tensile Strength*, ASTM International, 2007.

Cited in: Section 11.3

ASTM A500, *Standard Specification for Cold-Formed Welded and Seamless Carbon Steel Structural Tubing in Rounds and Shapes*, ASTM International, 2009.

Cited in: Section 14.1.3.3.4

ASTM A615, *Standard Specification for Deformed and Plain Carbon-Steel Bars for Concrete Reinforcement*, 2016.

Cited in: Section 14.2.4.5.5

ASTM A653, *Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process*, ASTM International, 2009.

Cited in: Section 14.1.3.3.3

ASTM A706/A706M, *Standard Specification for Low-Alloy Steel Deformed and Plain Bars for Concrete Reinforcement*, ASTM International, 2004.

Cited in: Section 14.2.4.5.5

ASTM C635/C635M, *Standard Specification for the Manufacture, Performance, and Testing of Metal Suspension Systems for Acoustical Tile and Lay-in Panel Ceilings*, ASTM International, 2013.

Cited in: Sections 13.5.6.2.1, 13.5.6.2.2

ASTM C636/C636M, *Standard Practice for Installation of Metal Ceiling Suspension Systems for Acoustical Tile and Lay-in Panels*, ASTM International, 2013.

Cited in: Sections 13.5.6.2.1, 13.5.6.2.2

ASTM C39, *Standard Test Method for Comprehensive Strength of Cylindrical Concrete Specimens*

Cited in: Section 14.2.4.6.8

ASTM C1087, *Standard Test Method for Determining Compatibility of Liquid-Applied Sealants with Accessories Used in Structural Glazing Systems*, ASTM International, 2011.

Cited in: Table 13.5-2

ASTM C1135, *Standard Test Method for Determining Tensile Adhesion Properties of Structural Sealants*, ASTM International, 2011.

Cited in: Table 13.5-2

ASTM C1184, *Standard Specification for Structural Silicone Sealants*, ASTM International, 2014.

Cited in: Table 13.5-2

ASTM C1265, *Standard Test Method for Determining the Tensile Properties of an Insulating Glass Edge Seal for Structural Glazing Applications*, ASTM International, 2011.

Cited in: Table 13.5-2

ASTM C1294, *Standard Test Method for Compatibility of Insulating Glass Edge Sealants with Liquid-Applied Glazing Materials*, ASTM International, 2011.

Cited in: Table 13.5-2

ASTM C1369, *Standard Specification for Secondary Edge Sealants for Structurally Glazed Insulating Glass Units*, ASTM International, 2014.

Cited in: Table 13.5-2

ASTM D1586, *Standard Test Method for Penetration Test and Split-Barrel Sampling of Soils*, ASTM International, 2004.

Cited in: Sections 11.3, 20.4.2

ASTM D2166, *Standard Test Method for Unconfined Compressive Strength of Cohesive Soil*, ASTM International, 2000.

Cited in: Section 20.4.3

ASTM D2216, *Standard Test Method for Laboratory Determination of Water (Moisture) Content of Soil and Rock by Mass*, ASTM International, 1998.

Cited in: Section 20.4.3

ASTM D2850, *Standard Test Method for Unconsolidated-Un drained Triaxial Compression Test on Cohesive Soils*, ASTM International, 2003.

Cited in: Sections 11.3, 20.4.3

ASTM D4318, *Standard Test Methods for Liquid Limit, Plastic Limit, and Plasticity Index of Soils*, ASTM International, 2000.

Cited in: Sections 11.3, 20.4.3

ASTM E580/E580M-14, *Standard Practice for Installation of Ceiling Suspension Systems for Acoustical Tile and Lay-In Panels in Areas Subject to Earthquake Ground Motions*, ASTM International, 2014.

Cited in: Sections 13.5.6.2.1, 13.5.6.2.2

ASTM F1554, *Standard Specification for Anchor Bolts, Steel, 36, 55, and 105-ksi Yield Strength*, ASTM International, 2015.

Cited in: Sections 13.5.3, 15.4.9.4

AWC NDS-15, *National Design Specification for Wood Construction*, Including Supplements, American Wood Council, 2014.

Cited in: Section 14.5.1

AWC SDPWS-15, *Special Design Provisions for Wind and Seismic*, American Wood Council, 2015.

Cited in: Sections 12.11.2.2.3, 12.14.7.2, 12.14.7.5.2, 14.5.1, Table 12.10-1

AWWA D100-11, *Welded Carbon Steel Tanks for Water Storage*, American Water Works Association, 2011.

Cited in: Sections 15.4.1, 15.7.7.1, 15.7.9.4, 15.7.10.6

AWWA D103, *Factory-Coated Bolted Carbon Steel Tanks for Water Storage*, American Water Works Association, 2009.

Cited in: Sections 15.4.1, 15.7.7.2, 15.7.9.5

AWWA D107, *Composite Elevated Tanks for Water Storage*, American Water Works Association, 2010.

Cited in: Section 15.7.10.7

AWWA D110, *Wire- and Strand-Wound, Circular, Prestressed Concrete Water Tanks*, American Water Works Association, 2013.

Cited in: Section 15.7.7.3

AWWA D115, *Tendon-Prestressed Concrete Water Tanks*, 2006.

Cited in: Section 15.7.7.3

DOC PS 1-09, *Structural Plywood*, U.S. Department of Commerce, National Institute of Standards and Technology, Gaithersburg, MD, 2009.

Cited in: Section 11.2

DOC PS 2-10, *Performance Standard for Wood-Based Structural-Use Panels*, U.S. Department of Commerce, National Institute of Standards and Technology, Gaithersburg, MD, 2011.

Cited in: Section 11.2

***FEMA P-795**, *Quantification of Building Seismic Performance Factors: Component Equivalency Methodology*. Applied Technology Council. Redwood City, CA, 2011.

***ICC-ES AC 156-04**, effective January 1, 2007, *Acceptance Criteria for Seismic Certification by Shake-Table Testing of Nonstructural Components and Systems*, International Code Council Evaluation Service, 2007.

Cited in: Section 13.2.5

***IRC**, *2015 International Residential Code*, International Code Council, 2012.

Cited in: Section 11.1.2

***MSS SP-58**, *Pipe Hangers and Supports—Materials, Design, and Manufacture*, Manufacturers Standardization Society of the Valve and Fittings, Industry, 2002.

Cited in: Section 13.6.4.1

NFPA 13, *Standard for the Installation of Sprinkler Systems*, National Fire Protection Association, 2016.

Cited in: Sections 13.4.6, 13.6.4.1, 13.6.7.2

NFPA 59A, *Standard for the Production, Storage, and Handling of Liquefied Natural Gas (LNG)*, National Fire Protection Association, 2013.

Cited in: Section 15.4.8

***NIST GCR 10-917-8**, *Evaluation of the FEMA P-695 Methodology for Quantification of Building Seismic Performance Factors*, NEHRP Consultants Joint Venture, a Partnership of the Applied Technology Council and the Consortium of Universities for Research in Earthquake Engineering. U.S. Department of Commerce, Washington, DC, 2010.

***NIST GCR 12-917-20**, *Tentative Framework for Development of Advanced Seismic Design Criteria for New Buildings*. NEHRP Consultants Joint Venture, a Partnership of the Applied Technology Council and the Consortium of Universities for Research in Earthquake Engineering. U.S. Department of Commerce, Washington, DC, 2012.

PTI DC10.5, *Standard Requirements for Design and Analysis of Post-Tensioned Concrete Foundations on Expansive Soils*, Post-Tensioning Institute, 2012.

Cited in: Section 12.13.9.2.1.1

TMS 402, *Building Code Requirements for Masonry Structures*, The Masonry Society, 2016.

Cited in: Sections 13.4.2.2, 14.4.1, 14.4.2, 14.4.3, 14.4.3.1, 14.4.4.1.1, 14.4.4.1.2, 14.4.5, 14.4.5.1, 14.4.5.2, 14.4.5.3, 14.4.5.4, 14.4.5.5, 14.4.6, 14.4.6.1, 15.4.9.2

TMS 602, *Specification for Masonry Structures*, The Masonry Society, 2016.

Cited in: Sections 14.4.1, 14.4.2, 14.4.7, 14.4.7.1

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